

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO 2 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
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Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Consider the concept map: Algorithm → Growth Function → Big-Omega → Lower Bound → Best-case Explain how lower bound represents minimum execution time. Extend the map with performance guarantee node. Justify why best-case analysis is useful. Interpret how this contributes to overall performance understanding.

Q2. Consider the concept map: Recurrence Relation → Substitution Method → Expansion → Simplification → Final Complexity Trace the step-by-step transformation of recurrence into a closed form. Extend the map with Guess and Verification nodes. Justify how substitution helps solve non-standard recurrences. Interpret how this structured approach improves understanding.

CO₂ - AT - 2 (WRITTEN)

Q - 132 Bubble sort: High Disorder Dataset

Dataset : [29, 11, 47, 6, 3, 14]

It compares adjacent elements and swaps them.

Pass 1 $\rightarrow$ [11, 29, 6, 3, 14, 47]

Pass 2 $\rightarrow$ [11, 6, 3, 14, 29, 47]

Pass 3 $\rightarrow$ [6, 3, 11, 14, 29, 47]

Pass 4 $\rightarrow$ [3, 6, 11, 14, 29, 47]

Pass 5 $\rightarrow$ No swaps, array remains [3, 6, 11, 14, 29, 47]

Total comparisons = $5 + 4 + 3 + 2 + 1 = 15$

Total swaps = $4 + 3 + 2 + 1 + 0 = 10$

best case = $O(n)$

average and worst case $O(n^2)$

Space = $O(1)$

$\rightarrow$ Bubble sort clearly demonstrates adjacent exchanges and is simple to implement, but repeated exchange

make it inefficient for large datasets

Q-31 selection sort

Dataset: $[44, 31, 44, 18, 9, 18]$

For descending order, selection sort repeatedly selects the maximum element from the unsorted position

Pass 1 Maximum = 44 $\rightarrow [44, 31, 44, 18, 9, 18]$

Pass 2 Maximum = 44 $\rightarrow [44, 44, 31, 18, 9, 18]$

Pass 3 Maximum = 31 $\rightarrow [44, 44, 31, 18, 9, 18]$

Pass 4 Maximum = 18 $\rightarrow [44, 44, 31, 18, 9, 18]$

Pass 5 Maximum = 18 $\rightarrow [44, 44, 31, 18, 18, 9]$

selection sort correctly handles repeated values and is appropriate when the objective is to repeatedly select the largest remaining element